Multiple Myeloma

81

Mohamad Mohty, Benedetto Bruno,
Nico Gagelmann, and Maria-Victoria Mateos

81.1 Which Patients Are Candidates for Autologous Hematopoietic Cell Transplantation?

The concept of high-dose therapy (HDT) followed by autologous hematopoietic cell transplantation (AHCT) remains the standard for treating newly diagnosed multiple myeloma in young and in select, fit, elderly patients. The introduction of ImiDs and proteasome inhibitors administered before and/or after HDT/AHCT gave way to the groundbreaking achievement of stringent complete response (sCR), immunophenotypic CR, and molecular CR, in addition to significantly increased CR and CR plus very good partial response rate (VGPR; Table 81.1). In randomized studies, age of participants is limited to 65 years to avoid selection bias and limit toxicities and withdrawal from studies. However, this does not mean that AAHCT is not feasible in older patients. A study whereby the median age of patients was 72 years old concluded that elderly multiple myeloma patients should not be excluded from transplantation displaying good results with melphalan 140 mg/m^2. Currently, in many centers, fit patients up to age 70, and even 75 years old, receive AHCT.

Renal impairment, per se, is not a contraindication to receiving HDT/AHCT with the recommendation for the use of 140 mg/m^2 of Melphalan. Nonetheless, it is a prompt reason to consider lower doses of therapy, as patients with renal impairment are more likely to suffer from HDM toxicities. Dialysis-dependent patients were more likely to develop toxicities and complications such as mucositis and infections, but had PFS and OS comparable to matched patients with normal renal function. Interestingly, a proportion of patients were able to attain dialysis-independence after transplantation.

M. Mohty (✉)
Department of Clinical Hematology and Cellular Therapy, Hospital Saint-Antoine, Sorbonne University, Paris, France
e-mail: mohamad.mohty@inserm.fr

B. Bruno
Department of Oncology, AOU Città della Salute e della Scienza, University of Torino, Torino, Italy

Department of Molecular Biotechnology and Health Sciences, AOU Città della Salute e della Scienza, University of Torino, Torino, Italy
e-mail: benedetto.bruno@unito.it

N. Gagelmann
Department of Stem Cell Transplantation, University Medical Center Hamburg-Eppendorf, Hamburg, Germany
e-mail: nico.gagelmann@posteo.de

M.-V. Mateos
University Hospital of Salamanca/IBSAL/Cancer Research Center-IBMCC (USAL-CSIC), Salamanca, Spain
e-mail: mvmateos@usal.es

© The Author(s) 2024
A. Sureda et al. (eds.), *The EBMT Handbook*, https://doi.org/10.1007/978-3-031-44080-9_81

Table 81.1 Current risk stratification and response criteria

Risk stratification
ISS: Stage I (beta2-m < 3.5 mg/L and albumin >3.3 g/dL), Stage III (beta2-m > 5.5 mg/L) and Stage II (all remaining cases).
1st revision (R-ISS): I: ISS I, standard-risk cytogenetics[a] and normal LDH, III: ISS III, plus high-risk cytogenetics or high LDH and II: all remaining cases.
2nd revision (R2-ISS): points assigned for ISS-III (1.5), ISS-II (1), del(17p) (1), high LDH (1), 1q + (0.5) I: 0 points II: 0.5–1 points III: 1.5–2.5 points IV: 3–5 points
Response criteria
MRD: bone marrow aspirate, analyze CD38, CD138, and CD45 in combination with CD19, CD56, CD27, CD81, and CD117 by FCM or sensitivity level $\leq 10^{-5}$ and 10^{-6} nucleated cells using next-generation sequencing **CR:** negative serum and urine immunofixation, less than 5% BMPCs and no soft-tissue plasmacytomas **Stringent CR:** as above plus normal free light-chain ratio and absence of clonal plasma cells. **VGPR:** 90% or more decrease in the serum M-protein and urine M-protein <100 mg/24 h. **PR:** 50% or more decrease in the serum M-protein, 90% or more decrease in urine M-protein or to <200 mg/24 h plus 50% or more decrease in soft-tissue plasmacytomas. **PD:** ≥1 of the following: increase in ≥25% from nadir in serum M-protein (absolute increase of at least 0.5 g/dL), urine M-protein (absolute increase ≥200 mg/24 h), BMPC (absolute increase ≥10%), soft-tissue plasmacytomas, development of new bone lesions, soft-tissue plasmacytomas, or hypercalcemia

MRD measurable residual disease, *CR* complete response, *VGPR* very good partial response, *PR* partial response, *PD* progressive disease, *BMPC* bone marrow plasma cells, *LDH* lactate dehydrogenase, *FCM* flow cytometry
[a]FISH identifying the following poor cytogenetic findings: t(4;14), t(14;16), and/or del 17p

81.2 What Is the Optimal Induction Treatment Prior to AHCT?

The role of induction is to decrease tumor burden, thus deepening the response rate and increasing the likelihood of engraftment, while retaining the maximum possible tolerability and minimum possible toxicity on normal hematopoietic cells. Multiple trials have proven the superiority of induction regimens containing one or two novel agents (thalidomide or bortezomib) over the historical VAD chemotherapy regimen in increasing CR, CR plus near-complete response (n-CR), or VGPR rates pre- and post-AHCT. Trials that compared two-drug (TD: thalidomide-dexamethasone or VD: bortezomib-dexamethasone) to three-drug induction (VTD: bortezomib, thalidomide, dexamethasone) have proven supremacy of the latter combination. VTD was also proven superior to bortezomib, cyclophosphamide and dexamethasone (VCD) thus highlighting the synergistic effect of combining an IMiD with bortezomib and dexamethasone. Furthermore, the use of 6 cycles of VTD (instead of 3–4 cycles) was associated with deeper responses. This is to be weighed against increased side effects, specifically neuropathy, upon administering 6 cycles instead of 3–4.

Similarly, the three-drug regimen bortezomib, lenalidomide, and dexamethasone (VRD) resulted in significantly increased PFS, response duration and OS resulting in the IFM introducing VRD as induction. In addition, the PETHEMA/GEM trial investigated induction with VRD-GEM with full dose lenalidomide from days 1 to 21, demonstrating an ORR of 85% post induction and 58% of patients achieving MRD-negativity post consolidation (Table 81.2).

Daratumumab (DARA), an anti-CD38+ monoclonal antibody, has been evaluated in patients with refractory disease. The Cassiopeia phase III trial and the Griffin phase II trial compared DARA-VTD to VTD and DARA-VRD to VRD respectively demonstrating positive results of adding daratumumab. Daratumumab plus cyclophosphamide, bortezomib, and dexamethasone (Dara-CyBorD) during induction was investigated in the phase II Lyra trial. Recent updates of the trial demonstrate activity and tolerability of Dara-CyBorD irrespective of high-risk cytogenetics with 12-month PFS and OS rates 87% and 99%, respectively. Finally, daratumumab is also being combined with carfilzomib, lenalidomide and dexamethasone (KRD) in a phase Ib trial whereby the combined regimen yielded 100%

Table 81.2 Selected prospective studies of AHCT vs no AHCT in newly diagnosed myeloma

	IFM 2009	BMTCTN 0702	Forte	Determination	Cardamon	EMN02/HO95
Study groups	LenBorDex + AHCT vs LenBorDex 8 cycles All with 1y Len maintenance	Tandem AHCT vs AHCT + LenBorDex vs AHCT All with Len maintenance until progression	CarLenDex + AHCT vs CarCyDex + AHCT vs CarLenDex 12 cycles Maintenance with CarLen or Len	LenBorDex vs LenBorDex + AHCT + LenBorDex Both groups with Len until disease progression	CarCyDex + AHCT vs CarCyDex 4 cycles All with Car maintenance 18 cycles	AHCT (single or tandem per center policy) vs VMP All with Len maintenance
Location	France	US	Italy	International (mainly USA)	UK	Europe
Follow-up, months	44	NR	51	76	40	60
Results						
PFS	Median: 50 months vs 36 months ($P < 0.001$)	38-months: 59% vs 58% vs 54%	4y: 69% vs 51% vs 56%	46 vs 68 months ($P < 0.001$)	2y: 75% vs 68%	Median:57 months vs 42 months ($P < 0.001$) 5y: 54% tandem AHCT vs 45% single AHCT
OS	3y: 81% vs 82%	6y: 73% vs 75% vs 76%	4y: 86% vs 76% vs 85%	5y: 79% vs 81%	2y: 94% vs 91%	5y: 80% tandem AHCT vs 73% single AHCT vs 72%

PFS progression-free survival, *OS* overall survival, *y* year, *AHCT* autologous hematopoietic cell transplantation, *Len* lenalidomide, *Dex* dexamethasone, *Bor* bortezomib, *Car* carfilzomib

ORR, 91% ≥ VGPR and 43% ≥ CR, with no negative impact on stem cell harvesting while retaining consistency of the DARA-KRD safety profile. Isatuximab is another anti-CD38 monoclonal antibody in investigation as part of the quadruplets combinations previously mentioned and there is a phase 3 GMMG-HD7 trial ongoing comparing Isa-RVd versus RVd before and after HDT-AHCT. Preliminary results showed a significantly higher undetectable MRD rate in the Isa-RVd arm (50.1%) versus RVd alone (35.6%) after 3 × 6-week cycles.

MRD negativity, defined as the absence of disease within 10^5 bone marrow cells, has been examined due to its important prognostic value at different stages of the transplantation process. The final analysis of the IFM2009 prospective trial demonstrated the significance of MRD negativity whereby patients achieving MRD negativity after induction with VRD had a similar OS irrespective of whether they received an AHCT or not. Similar results were reported in the phase 3 Determination study, with a similar design to the IFM2009 but maintenance with lenalidomide was given until progression disease. The lack of benefit in OS was equally observed and, in this study, only 29% of the patients in the control arm received AHCT at relapse (89% in the IFM2009 trial). In addition, MRD negativity proved in both studies a more powerful predictor of outcome than cytogenetics whereby patients with high-risk cytogenetics who achieved MRD negativity had better outcomes than patients with standard-risk cytogenetics who did not. Thus, MRD could potentially become essential in stratifying patients during maintenance and consolidation randomization and when deciding on maintenance duration.

81.3 What Is the Optimal Conditioning Regimen Prior to AHCT?

The current accepted standard for HDT is intravenous high-dose melphalan (200 mg/m^2). Previous trials attempting to replace this with oral and intravenous busulfan have failed, due to increased toxicity and lack of superiority.

Similarly, studies failed to show a significant benefit for combining bortezomib and high-dose melphalan. The role of bendamustine added to melphalan is being explored and not well established yet. As such, HDM remains the standard conditioning regimen prior to AHCT awaiting results of clinical trials of other conditioning regimens (if any).

81.4 What Is the Impact of Consolidation Therapy After AHCT?

The second randomization in the EMN02/HO95 trial compared the aftermaths of receiving 2 cycles of VRD consolidation followed by lenalidomide versus lenalidomide maintenance alone, demonstrating the significant advantage VRD consolidation inferred in prolonging PFS. Moreover, PFS was prolonged in most of the predefined groups in the study including ISS I and II, low-risk cytogenetics, irrespective of whether patients received VMP (bortezomib, melphalan and prednisone) or transplantation prior to consolidation. Nonetheless, VRD consolidation failed to improve PFS in patients with high-risk cytogenetics ((del(17p) and/or t(4;14) and/or t(14;16)). This confirms the benefit of VRD consolidation followed by lenalidomide maintenance in younger, newly diagnosed multiple myeloma patients with low-risk disease. Along the same line of the EMN02/HO95 trial, the StaMINA phase III trial randomized patients to compare HDM/AHCT plus VRD consolidation plus lenalidomide maintenance, versus tandem HDM/AHCT plus lenalidomide maintenance, versus single HDM/AHCT plus lenalidomide maintenance. It concluded that the addition of VRD consolidation or a tandem AHCT was not superior to standard AHCT followed by lenalidomide in upfront treatment of newly diagnosed multiple myeloma. With the currently available data, the role of post-transplant consolidation remains controversial. European guidelines do not make any formal recommendation about consolidation, but the most recent phase 3 trials that incorporate the

anti-CD38 monoclonal antibodies as part of the induction with either VTD or VRD do include 2 cycles of consolidation after 4 induction cycles followed by AHCT. A practical possibility not evaluated in prospective trials would be to consider the consolidation as optional depending of the depth of response achieved after AHCT, and it would not be necessary for those patients already in CR and undetectable MRD.

81.5 What Is the Impact of Maintenance Therapy After AHCT?

AHCT is not curative, and progressions and relapses are common even if CR is attained post-transplant. Maintenance therapy is thus added and is expected to be gentle with the safest profile post AHCT, but unlike consolidation, it is administered long-term to deepen the response, prevent progression, and prolong OS. Thalidomide maintenance demonstrated benefit in terms of response rates but not OS and was repeatedly associated with peripheral neuropathy, fatigue, and other side effects, all of which resulted in patient-reported decreased quality of life despite prolonged duration of disease control. In contrast, lenalidomide maintenance has been shown to be well tolerated and to dramatically improve PFS and OS. A meta-analysis of 3 randomized trials, CALGB, IFM, and GIMEMA, that compared lenalidomide maintenance to placebo or observation, has demonstrated a significantly improved PFS in all subgroups of patients regardless of age, myeloma severity and staging, and induction regimen (52.8 versus 23.5 months), even though patients who had received lenalidomide in induction, or had achieved a deeper response post-transplant, were more likely to benefit from lenalidomide. OS was also significantly improved in the lenalidomide arm, except in women older than 60 years with poor cytogenetics. Overall, the addition of lenalidomide reduced the chance of death by a substantial 25%, thus increasing median survival by approximately 2.4 years. As demonstrated in previous studies, an increased incidence of second primary malignancies, albeit modest, was associated with lenalidomide, though the time to death due to a second primary malignancy did not differ between the two groups. Such results propose lenalidomide as a standard maintenance drug in transplant-eligible patients. Recent updates of the Myeloma XI trial's results were in concordance with the meta-analysis.

So far in previous trials, lenalidomide has been given in low doses until progression or adverse events develop, and this practice is currently approved by both, FDA and EMA. Given that 30% of cases with premature termination of lenalidomide were attributed to toxicities and second primary malignancies, the question that remains is regarding the optimal duration of treatment with lenalidomide for safety and cost.

Finally, bortezomib was also tested as part of maintenance, either alone or in combination with IMiDs, demonstrating improved PFS, but not OS. Nonetheless, bortezomib poses an obstacle due to its subcutaneous administration. The first oral PI, ixazomib, has been evaluated as maintenance in comparison with placebo in a phase 3 trial and although the trial met its primary endpoint, the magnitude of the benefit is limited with a difference in PFS of only 5 months (26.5 vs 21.3) and not approved in this setting. Monoclonal antibodies anti-CD38 are being evaluated as maintenance, in majority of the trials in combination with lenalidomide, and there are at least two phase 3 clinical trials comparing lenalidomide with lenalidomide plus daratumumab and one comparing lenalidomide with lenalidomide plus isatuximab. Results are awaited to confirm the superiority of such combination.

81.6 What Is the Value of Single Versus Tandem AHCT?

The concept of tandem AHCT came about in an era where conventional chemotherapy was the only available drug. Previous randomized trials had demonstrated improved outcomes with tandem transplantation in terms of PFS and OS even in patients who had not achieved a VGPR after

the first transplant. An alternative treatment approach, total therapy 3 (TT3), including induction, tandem AHCT, consolidation, and maintenance, has allowed one of the best results to be achieved (CR/nCR rate of 83%, 2-year PFS of 84%, and 2-year OS of 86%).

In the modern era, the impact of tandem transplantation was evaluated in the EMN02/HO95 and StaMINA trials. The EMN02/H095 trial explored the result of tandem versus single transplantation in newly diagnosed multiple myeloma patients. Tandem transplantation was shown to improve the depth of the response by 25% with more than 50% of the patients achieving at least a CR. PFS and OS were significantly improved after a second transplant, with approximately 30% reduction in the risk of death and progression. Updated results of the EMN02/HO95 confirmed the improved 3-year PFS from 63 months after one AHCT to 73.1 after two AHCTs. Importantly, the positive effect of tandem AHCT was seen in high-risk groups, in which randomization to receive double AHCT was found to be an independent predictor of PFS. The analysis thus concluded that double frontline AHCT was superior to single AHCT in terms of PFS and OS in all patients, including poor prognosis subgroups, indicating that the latter were the most likely to benefit. On the other hand, the StaMINA trial failed to show superiority of tandem versus single transplant in the era of novel agents. It is noteworthy that more than 30% of patients randomized to tandem transplant did not receive the second transplant.

Overall, with the currently available data, a second AHCT may be beneficial in high-risk patients including patients with high-risk cytogenetics and R-ISS 3 category of disease.

The next challenge is to evaluate the necessity of HDT/AHCT when a monoclonal antibody such as daratumumab or isatuximab is added to a powerful induction regimen combining an IMiD and a PI, and whether this strategy can cure a fraction of patients. This strategy is currently being evaluated in at least 2 different phase 3 clinical trials (MIDAS and MASTER-2). The impending challenge remains whether or not transplantation will be later substituted by less intensive novel agent combinations.

81.7 What Is the Role of AHCT as Salvage Therapy?

Salvage therapy is defined as AHCT given to a patient with signs of disease progression after an earlier AHCT. By the BSBMT/UKMF Myeloma X trial, salvage AHCT with 200 mg/m^2 melphalan was superior to cyclophosphamide 400 mg/m^2 weekly for 12 weeks upon relapse and reinduction with VAD. The time to disease progression (19 versus 11 months) and OS (67 vs. 52 months) were significantly in favor of salvage AHCT. As such, AHCT can be considered for salvage in fit patients if the interval between the first AHCT and relapse is relatively very long (e.g. at least 2–3 years). The randomized GMMG phase III trial ReLApsE compared salvage AHCT and lenalidomide maintenance versus. Lenalidomide/dexamethasone for relapsed multiple myeloma and although the incorporation of salvage AHCT into relapse treatment with Rd was feasible in 71% of patients it did not significantly prolong PFS and OS on ITT analysis. Overall, and except in a few highly selected patients, the role of AHCT in the relapsed setting is currently questionable.

81.8 What Is the Role of Allogeneic Hematopoietic Cell Transplantation in Multiple Myeloma?

Allogeneic transplantation may result in durable remissions in subsets of myeloma patients. However, a long follow-up is required to observe a potential survival benefit after passing the initial risk of transplant-related mortality. Trials that employed the biologic assignment based on the availability of a HLA-matched sibling donor reported conflicting results in newly diagnosed myeloma. Two trials reported better PFS and OS with an “allo-approach”. However, a meta-analysis showed that the benefits of allografting are offset by high rates of treatment-related mortality. Currently, due to the lack of robust clinical data, there is considerable uncertainty, in the era

Table 81.3 Results of the key CAR-T cell studies in multiple myeloma

Product	N	Lines of therapy, median	High-risk cytogenetics/ EMD	ORR	PFS, median in months	CRS/ICANS (grades 3–4)
Ide-cel	128	6	35%/39%	73%	9	5%/3%
	386	3	42%/24%	71%	13	4%/3%
Cilta-cel	97	6	24%/13%	98%	35	4%/9%
	419	2	59%/21%	85%	NR	1%/3%
ARI0002h	30	4	33%/20%	100%	15	0%/0%

of novel agents and consolidation/maintenance strategies concerning the subset of patients who could most benefit from an allograft and its timing. Novel strategies could not be designed without the combination of new drugs that may enhance graft-versus-myeloma effects to allow long-term disease control and prolong OS, in particular in patients with high-risk disease. Enhancement of the graft-versus-myeloma effect after allografts through the immunomodulatory properties of novel agents may be a key factor to improve clinical outcomes. There is evidence that a synergy between the donor immune system and immunomodulatory drugs may prolong response. Though optimal timing of an allograft and dosage of new drugs remain to be determined, in the absence of other alternatives, younger patients with high-risk features, suboptimal response, and/or early relapse may be considered by some investigators for an allograft, preferably in the context of a clinical trial.

81.9 The Emerging Role CAR-T Cells in Multiple Myeloma

Chimeric antigen receptor (CAR-) T cell therapy represents a new and promising option for relapsed/refractory myeloma patients. Two commercial products, both targeting the myeloma-specific antigen B cell maturation antigen (BCMA), have been approved: idecabtagene vicleucel (ide-cel) and ciltacabtagene autoleucel (cilta-cel). The indications for both products are patients with >3 previous lines of therapy, including an IMiD, proteasome inhibitors, and anti-CD38 monoclonal antibodies. Subsequent phase 3 results have been reported. The KarMMa-3 study enrolled participants with 2–4 previous lines, randomizing ide-cel or standard-of-care chemotherapy regimens. The median PFS was 13 months in the ide-cel group versus 4 months. The CARTITUDE-4 trial included patients with lenalidomide-refractory MM and 1–3 previous lines. The median PFS was not reached for the cilta-cel group versus 12 months. Remarkably, these therapies also induced deep responses in high-risk patients, including those with extramedullary disease, highlighting its potential to revolutionize treatment algorithms for these patients.

Because of the financial burden of such therapies, academic CAR-T products are emerging. ARI0002h (an academic product from Spain) was administered in a fractioned manner with a booster dose after 3 months for patients after >1 previous lines of therapy including a proteasome inhibitor, an IMiD, and an anti-CD38 antibody. This induced deep and sustained responses, with a low toxicity. Other efforts from Israel achieved comparable results, underscoring the potential for point-of-care CAR-T cell therapy that may increase accessibility without loss of efficacy (Table 81.3).

81.10 Conclusion and Future Perspectives

More than thirty years after its introduction, HDT/AHCT remains the standard of care for patients with newly diagnosed multiple myeloma. Despite the advent of novel agents, AHCT remains a very common treatment modality, especially in Europe and is included in all ongoing and proposed trials. The latter is yet to be challenged by many novel agents (including earlier use of CAR T-cells) which are continuously

explored. When dealing with unfavorable cytogenetics and poor prognostic factors, tandem transplantation appears to be an encouraging strategy. Whereas the use of post-transplant consolidation is controversial, lenalidomide maintenance prolongs PFS and OS and can be considered as a standard of care. One important endpoint to be measured in studies is MRD negativity in standard or high-risk disease, since it is a significant predictor of PFS, OS, and potential cure in a fraction of patients.

Key Points

- AHCT is the preferred treatment approach (standard of care) in young and fit myeloma patients.
- Prior to AHCT, patients should receive at least a triplet-based induction regimen (ideally a quadruplet including an anti-CD38 antibody) aiming to achieve a deep response.
- High-dose MEL 200 mg/m^2 is the standard conditioning for AHCT in myeloma.
- Patients should receive some form of post AHCT therapy (maintenance therapy preceded eventually by consolidation).
- Double AHCT can be considered for high-risk myeloma (e.g., patients with a del17p cytogenetic abnormality).
- The role of allogeneic-HCT (briefly discussed in this chapter) is highly controversial in myeloma and should be performed as part of a clinical trial.
- In the mid-term, the role of AHCT will be challenged by the advent of CAR-T cells.

Further Reading

Cavo M, Tacchetti P, Patriarca F, Petrucci MT, Pantani L, Galli M, et al. Bortezomib with thalidomide plus dexamethasone compared with thalidomide plus dexamethasone as induction therapy before, and consolidation therapy after, double autologous stem-cell transplantation in newly diagnosed multiple myeloma: a randomised phase 3 study. Lancet. 2010;376(9758):2075–85.

Durie BG, Hoering A, Abidi MH, Rajkumar SV, Epstein J, Kahanic SP, et al. Bortezomib with lenalidomide and dexamethasone versus lenalidomide and dexamethasone alone in patients with newly diagnosed myeloma without intent for immediate autologous stem-cell transplant (SWOG S0777): a randomised, open-label, phase 3 trial. Lancet. 2017;389(10068):519–27.

Facon T, Mary JY, Hulin C, Benboubker L, Attal M, Pegourie B, et al. Melphalan and prednisone plus thalidomide versus melphalan and prednisone alone or reduced-intensity autologous stem cell transplantation in elderly patients with multiple myeloma (IFM 99-06): a randomised trial. Lancet. 2007;370(9594):1209–18.

Harousseau JL, Moreau P. Autologous hematopoietic stem-cell transplantation for multiple myeloma. N Engl J Med. 2009;360(25):2645–54.

Harousseau JL, Attal M, Avet-Loiseau H, Marit G, Caillot D, Mohty M, et al. Bortezomib plus dexamethasone is superior to vincristine plus doxorubicin plus dexamethasone as induction treatment prior to autologous stem-cell transplantation in newly diagnosed multiple myeloma: results of the IFM 2005-01 phase III trial. J Clin Oncol. 2010;28(30):4621–9.

Kharfan-Dabaja MA, Hamadani M, Reljic T, et al. Comparative efficacy of tandem autologous versus autologous followed by allogeneic hematopoietic cell transplantation in patients with newly diagnosed multiple myeloma: a systematic review and meta-analysis of randomized controlled trials. J Hematol Oncol. 2013;6:2.

Koreth J, Cutler CS, Djulbegovic B, Behl R, Schlossman RL, Munshi NC, et al. High-dose therapy with single autologous transplantation versus chemotherapy for newly diagnosed multiple myeloma: a systematic review and meta-analysis of randomized controlled trials. Biol Blood Marrow Transplant. 2007;13(2):183–96.

McCarthy PL, Holstein SA, Petrucci MT, Richardson PG, Hulin C, Tosi P, et al. Lenalidomide maintenance after autologous stem-cell transplantation in newly diagnosed multiple myeloma: a meta-analysis. J Clin Oncol. 2017;35(29):3279–89.

Mohty M, Harousseau JL. Treatment of autologous stem cell transplant-eligible multiple myeloma patients: ten questions and answers. Haematologica. 2014;99(3):408–16.

Moreau P, Hulin C, Perrot A, Arnulf B, Belhadj K, Benboubker L, et al. Maintenance with daratumumab or observation following treatment with bortezomib, thalidomide, and dexamethasone with or without daratumumab and autologous stem-cell transplant in patients with newly diagnosed multiple myeloma (CASSIOPEIA): an open-label, randomised, phase 3 trial. Lancet Oncol. 2021;22(10):1378–90.

Palumbo A, Triolo S, Argentino C, Bringhen S, Dominietto A, Rus C, et al. Dose-intensive melphalan with stem cell support (MEL100) is superior to stan-

dard treatment in elderly myeloma patients. Blood. 1999;94(4):1248–53.
Perrot A. How I treat frontline transplantation-eligible multiple myeloma. Blood. 2022;139(19):2882–8.
Richardson PG, Jacobus SJ, Weller EA, Hassoun H, Lonial S, Raje NS, et al. Triplet therapy, transplantation, and maintenance until progression in myeloma. N Engl J Med. 2022;387(2):132–47.
Sonneveld P, Schmidt-Wolf IG, van der Holt B, El Jarari L, Bertsch U, Salwender H, et al. Bortezomib induction and maintenance treatment in patients with newly diagnosed multiple myeloma: results of the randomized phase III HOVON-65/ GMMG-HD4 trial. J Clin Oncol. 2012;30(24):2946–55.
Sonneveld P, Goldschmidt H, Rosinol L, Blade J, Lahuerta JJ, Cavo M, et al. Bortezomib-based versus nonbortezomib-based induction treatment before autologous stem-cell transplantation in patients with previously untreated multiple myeloma: a meta-analysis of phase III randomized, controlled trials. J Clin Oncol. 2013;31(26):3279–87.
Stadtmauer EA, Pasquini MC, Blackwell B, Hari P, Bashey A, Devine S, et al. Autologous transplantation, consolidation, and maintenance therapy in multiple myeloma: results of the BMT CTN 0702 trial. J Clin Oncol. 2019;37(7):589–97.
Yakoub-Agha I, Einsele H. The EBMT/EHA CAR-T cell handbook. In: Kröger N, Gribben J, Chabannon C, Yakoub-Agha I, Einsele H, editors. Multiple myeloma. 1st ed. Cham: Springer International Publishing; Imprint Springer; 2022. 237 p.

Open Access This chapter is licensed under the terms of the Creative Commons Attribution 4.0 International License (http://creativecommons.org/licenses/by/4.0/), which permits use, sharing, adaptation, distribution and reproduction in any medium or format, as long as you give appropriate credit to the original author(s) and the source, provide a link to the Creative Commons license and indicate if changes were made.

The images or other third party material in this chapter are included in the chapter's Creative Commons license, unless indicated otherwise in a credit line to the material. If material is not included in the chapter's Creative Commons license and your intended use is not permitted by statutory regulation or exceeds the permitted use, you will need to obtain permission directly from the copyright holder.

